

Basic Information

Experiment No.: 07

Date: 24-09-2026

Title: Java Stack and Its Methods

Objective / Aim

To implement the "Stack" class in Java and demonstrate stack operations such as "push()", "peek()", "search()", "pop()", and "empty()".

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

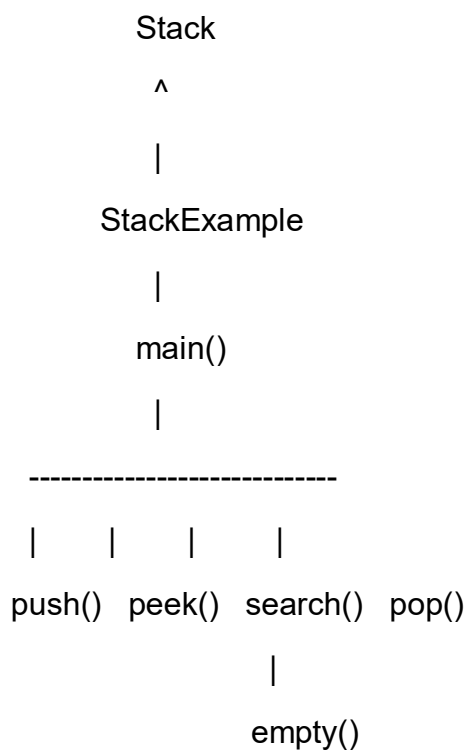
- Java JDK
 - Any Java IDE or Command Prompt
 - Operating System: Windows/Linux/macOS
-

Algorithm / Logic Description

1. Create a "Stack" of strings.
2. Add elements to the stack using "push()".
3. Display the stack.
4. View the top element using "peek()".
5. Find an element's position using "search()".

6. Remove the top element using "pop()".
 7. Display the stack after removal.
 8. Check whether the stack is empty using "empty()".
 9. End the program.
-

UML Class Diagram



Source Code

```
package collectionframework;
import java.util.Stack;

public class StackExample {
    public static void main(String[] args) {
```

```
Stack<String> stack = new Stack<>();

// push()
stack.push("Apple");
stack.push("Banana");
stack.push("Mango");

System.out.println("Stack: " + stack);

// peek()
System.out.println("Top element: " + stack.peek());

// search()
System.out.println("Position of Banana: " +
    stack.search("Banana"));

// pop()
System.out.println("Popped element: " + stack.pop());

System.out.println("Stack after pop: " + stack);

// empty()
System.out.println("Is Stack Empty: " + stack.empty());
}
}
```

Compilation & Execution Commands

Save the file as:

StackExample.java

Since the program contains:

package collectionframework;

the file should be inside the "collectionframework" folder.

Compile:

javac StackExample.java

Run:

java collectionframework.StackExample

Output Screen

Stack: [Apple, Banana, Mango]

Top element: Mango

Position of Banana: 2

Popped element: Mango

Stack after pop: [Apple, Banana]

Is Stack Empty: false

Result

The "Stack" class was successfully implemented, and the "push()", "peek()", "search()", "pop()", and "empty()" operations were successfully demonstrated.